CodeAlpha

Task 2 — Exploratory Data Analysis

EDA of the scraped books dataset

Source file: books_scraped.csv
Dataset size: 100 records × 4 original fields

Prepared as a complete EDA submission covering data structure, meaningful questions, descriptive statistics, trends, anomalies, hypotheses, visual analysis, and data-quality issues.

1. Executive Summary

- The dataset contains **100 books** with four scraped fields: Title, Price, Availability, and Rating.
- All 100 records are marked **In stock**, so availability has no variation in this sample.
- Book prices range from **£10.16** to **£58.11**, with a mean of **£34.56** and median of **£34.78**.
- Ratings are distributed across all five levels. The most frequent ratings are One and Three, with **22** and **22** books respectively.
- The Pearson correlation between numeric rating and price is **-0.122**, indicating a very weak negative linear relationship in this sample.
- Using the 1.5×IQR rule, **0 price outliers** are detected. The dataset therefore shows a broad price spread but no statistical price outliers by this standard.

Key takeaway

The scraped sample is useful for understanding price and rating distributions, but it is limited for deeper business analysis because it lacks category, author, publication date, review count, and other explanatory variables. The absence of availability variation also prevents meaningful stock-status comparisons.

100	£34.56	2.93/5	-0.122
Books	Mean price	Mean rating	Price–rating correlation

2. Data Structure and Variables

The original CSV contains four columns. For analysis, Price was converted from a currency-formatted string into a numeric GBP variable, and Rating was converted from words (One–Five) to an ordinal numeric scale from 1 to 5.

Variable	Original type	EDA type	Role / interpretation
Title	object/text	categorical text	Book identifier / descriptive label
Price	object/text	numeric (GBP)	Continuous quantitative variable
Availability	object/text	categorical	Stock status
Rating	object/text	ordinal + numeric	1–5 ordered rating scale

Data completeness

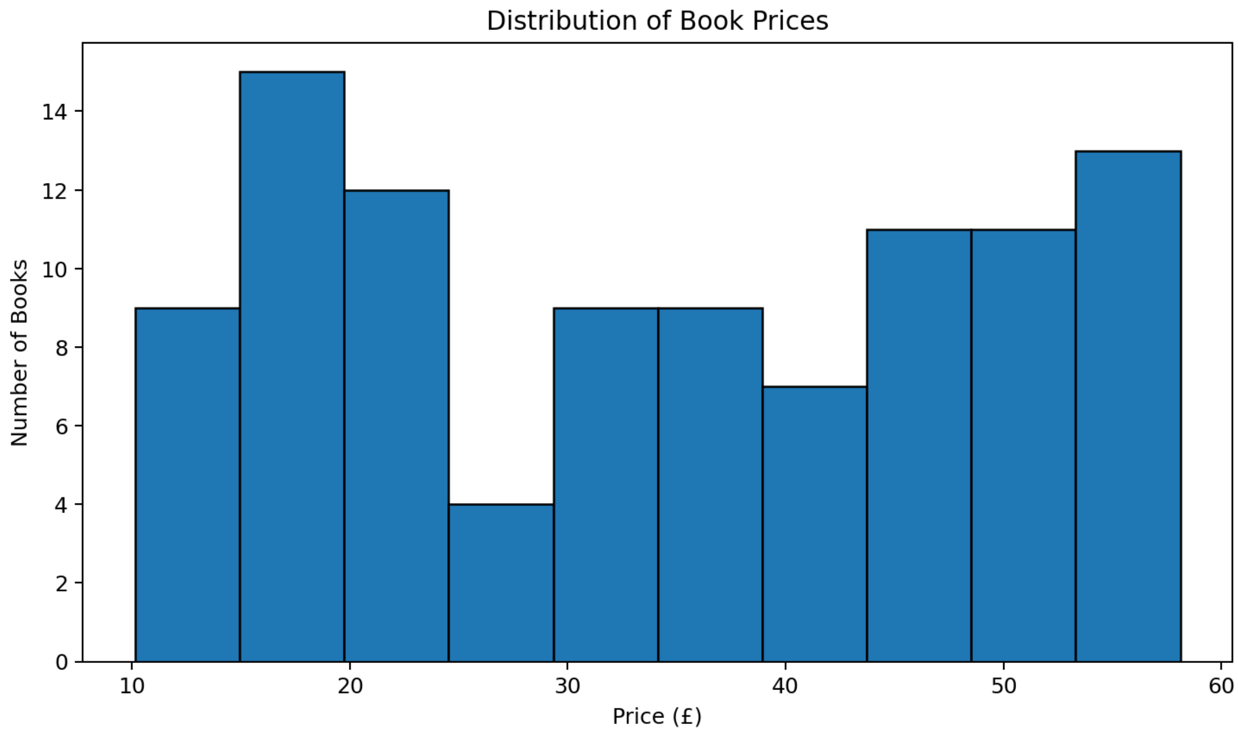
Check	Result
Rows	100
Original columns	6
Missing values	0 in every original column
Unique titles	100
Availability values	In stock
Rating values	One, Two, Three, Four, Five

Meaningful questions for analysis

- What does the distribution of book prices look like, and how wide is the price range?
- Are the five rating levels represented evenly?
- Is there evidence that higher-rated books tend to be cheaper or more expensive?
- Are there unusual prices that may require investigation?
- Does availability provide any useful segmentation in this dataset?

3. Price Analysis

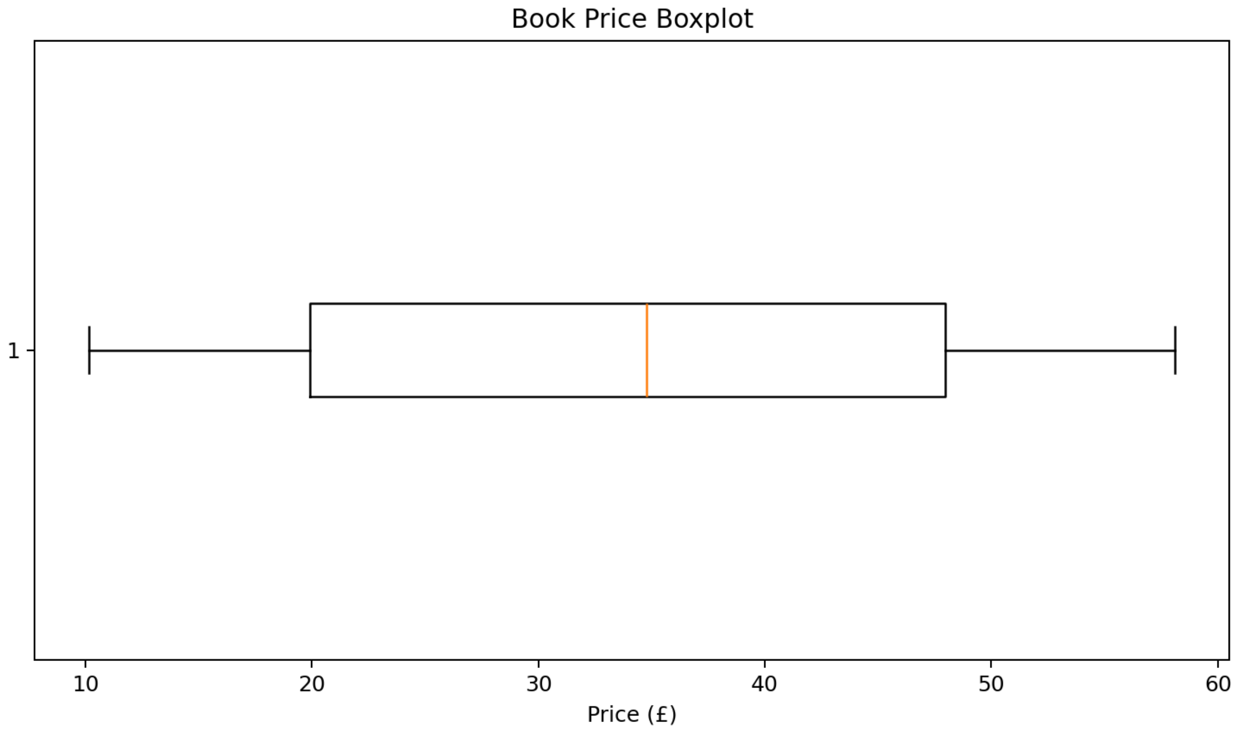
Prices span **£10.16–£58.11**. The middle 50% of prices lies between **£19.90** and **£47.97**. The mean (£34.56) is very close to the median (£34.78), suggesting that the central tendency is not being strongly distorted by a small number of extreme observations.



The histogram shows a broad spread across the available price range rather than a narrow cluster. Prices are distributed across multiple bands, which makes price a useful variable for descriptive EDA.

4. Outlier and Anomaly Review

Using the standard 1.5×IQR rule, Q1 = £19.90, Q3 = £47.97, and IQR = £28.07. The resulting fences are £-22.21 and £90.07. Because all observed prices fall within these limits, **no price observations are statistical outliers under the IQR rule**.



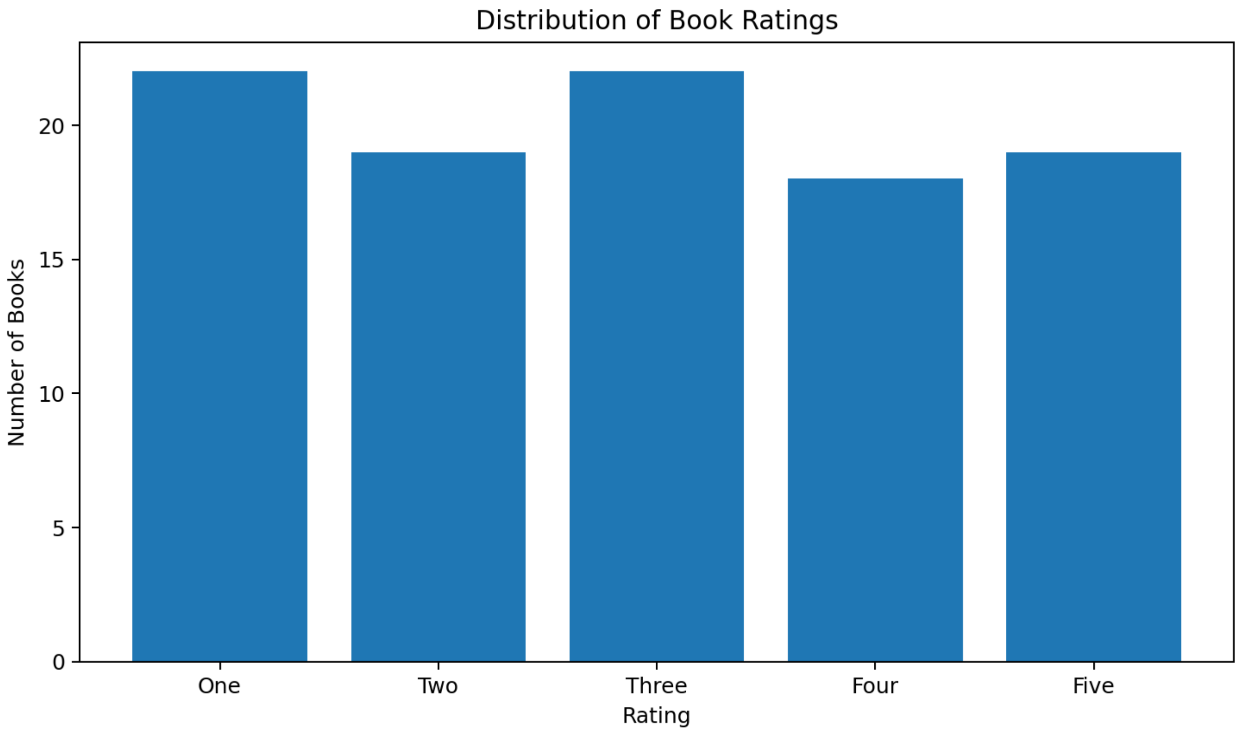
Highest-priced books

Book	Price (£)	Rating
The Death of Humanity: and the Case for Life	58.11	Four
Slow States of Collapse: Poems	57.31	Three
Our Band Could Be Your Life: Scenes from the America...	57.25	Three
The Past Never Ends	56.50	Four
The Pioneer Woman Cooks: Dinnertime: Comfort Classic...	56.41	One

Important distinction: absence of IQR outliers does not mean every record is error-free. Extremely high or low prices may still deserve source-level verification if the business context suggests they are implausible.

5. Rating Analysis

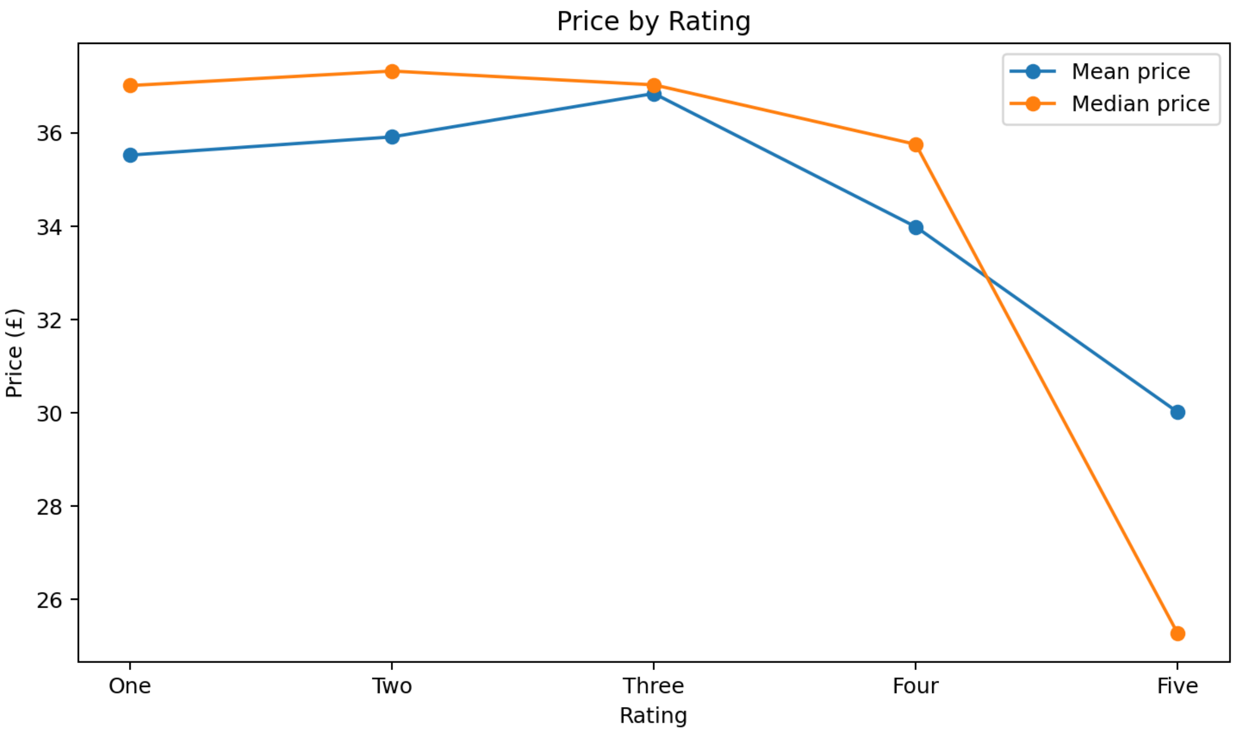
The average rating is **2.93/5**, while the median rating is **3/5**. All five rating categories occur, with counts ranging from 18 to 22 books. This is a relatively balanced rating distribution, so no single rating dominates the sample.



Rating	Count	Share
One	22	22.0%
Two	19	19.0%
Three	22	22.0%
Four	18	18.0%
Five	19	19.0%

6. Relationship Between Price and Rating

To test whether rating is associated with price, the word ratings were encoded from 1 to 5 and compared with Price_GBP. The Pearson correlation is **-0.122**, which is close to zero and indicates only a very weak negative linear association.



The grouped comparison also shows that the mean price does not increase consistently with rating. In this sample, the five-star group has a lower average price than the one-, two-, and three-star groups. This is an observational pattern, not evidence that rating causes price differences.

7. Hypotheses, Assumptions, and Validation

Hypothesis 1 — Higher-rated books have higher prices

Result: Not supported by the observed data. The correlation is -0.122, and the mean price does not rise monotonically from rating 1 to rating 5. A larger dataset with more explanatory variables would be needed for a stronger conclusion.

Hypothesis 2 — The rating distribution is highly imbalanced

Result: Not supported. Counts are [22, 19, 22, 18, 19], so the five rating levels are reasonably represented.

Hypothesis 3 — The scraped sample contains obvious price outliers

Result: Not supported under the IQR criterion. No price falls outside the calculated lower and upper fences.

Assumptions used

- Rating words represent an ordered five-point scale: One < Five.
- Price values are expressed in GBP and the currency symbol can be removed without changing the numeric value.
- Pearson correlation is used as a descriptive measure of linear association, not as proof of causation.
- The scraped rows are treated as the available sample and are not assumed to represent every book in the source population.

8. Data Issues, Limitations, and Recommendations

Potential data issues

- The dataset contains only four original fields, limiting the ability to explain why prices or ratings differ.
- Availability is constant ('In stock' for all rows), so stock-status analysis has no variability.
- Price and rating arrived as text and require explicit transformation before statistical analysis.
- No category, author, publication date, review count, or review text is available, preventing category-level and sentiment-oriented analysis.
- The dataset does not include a scrape timestamp or source URL per record, which makes provenance and reproducibility weaker.

Recommended next steps

- Scrape additional fields such as category, author, product URL, number of reviews, and publication metadata if available.
- Capture multiple scraping dates to analyze price changes and availability changes over time.
- Validate currency and numeric conversions against the source page format.
- If review counts or review text are collected, investigate whether popularity or sentiment is associated with price and rating.
- For inferential modeling, increase the sample size and consider regression or non-parametric tests after checking assumptions.

Conclusion

This EDA establishes a clean baseline for the 100-book scraped dataset. Prices vary substantially from £10.16 to £58.11, ratings are broadly balanced, and there is no meaningful linear price–rating relationship in the sample. The strongest limitation is not a statistical anomaly but missing context: additional book-level variables are needed to explain the observed differences and support more advanced analysis.

Submission artifact: This PDF is designed as a standalone Task 2 EDA report suitable for inclusion in a GitHub project repository.